



Barangay Vehicle Management System

System Capabilities & Workflow Reference

A smarter way to manage a barangay's emergency fleet —
availability, maintenance, and readiness.

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Laravel API backend · React single-page frontend

1. What this system is for

The VMS keeps a barangay's **emergency vehicles** — ambulances, fire trucks, and rescue boats — **ready when they are needed**. It answers two questions at all times:

- **Availability** — can this vehicle be dispatched right now?
- **Maintenance** — what is being repaired, by whom, at what stage, and is it proven to work before it goes back into service?

Deliberately out of scope: trip/operations logging, mileage tracking, and government document-compliance. The system tracks *readiness and repair*, not dispatch or paperwork.

Two independent facts about every vehicle. *Status* = availability (Available / Under Maintenance / Inactive / Decommissioned). *Condition* = physical state (Good / Needs Inspection / Needs Repair / Damaged). They are tracked separately — a vehicle can be Available but flagged Needs Inspection, or pulled Under Maintenance while its condition is being fixed.

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3. Roles & multi-role accounts

The system has three roles. Each is a set of permissions — a "hat" — not a headcount.

Role	What they do
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Admin	Registers vehicles & users, creates tickets, assigns mechanics, gives the final confirmation, closes/cancels/decommissions, runs reports.
Custodian	Reports vehicle issues, performs assigned inspections, and runs the functional test that accepts a repair back into service (the "user" who confirms it works).
Maintenance Personnel	Receives work orders, performs and logs repairs (parts, cost, photo/document), and submits the work for verification.

Multi-role support. One account can hold *several* roles at once (e.g. a small barangay's single staffer is both Custodian and Maintenance). The account keeps a *primary* role for its portal, but the sidebar shows the *union* of every module across its roles, and permission checks accept any hat the person holds. Every action still records *which role* performed it, so accountability survives even when one person does several steps. Separation of duties is enforced by **role**, not by person — so the same system fits a two-person barangay and a twenty-person one with no changes.

4. Vehicle management (Land & Water)

Admins register and maintain every vehicle. Each vehicle belongs to a **Vehicle Type** (category) that carries a **domain** — **Land** or **Water** — which changes the specs the form asks for.

Domain	Specialised spec fields	Capacity units
Land (ambulance, fire truck)	Fuel Type	kg, tons, L, pax
Water (rescue boat)	Hull Material (Fiberglass / Aluminum / Steel / Wood / Rubber-Inflatable), Engine Type, Fuel Type	L, gal, m ³ , pax

Capabilities: add / edit / photo upload / archive (temporary, reversible) / **decommission** (permanent end-of-life, see §10) / restore. Every change is written to the vehicle's history and the system activity log.

5. Issue reporting

A Custodian (or Admin) reports a vehicle problem — type, severity, description, optional photo. Reports move through **Pending → Under Review → In Maintenance → Resolved**. An unresolved report flags the vehicle's condition so it surfaces on the dashboard. A report can be escalated by an Admin into a maintenance ticket.

Retired vehicles (archived or decommissioned) cannot receive new issue reports or tickets — they are out of the fleet.

6. The maintenance ticket workflow

The heart of the system. A **ticket** is a **Main Issue** container (e.g. "Overheating") that holds one or more **Sub-Issues** (root causes, e.g. "low coolant level", "faulty radiator"). Each sub-issue runs its *own* independent repair pipeline with its own mechanic, so a ticket's progress reads as **X of N done**.

Ticket lifecycle

Open → Active → Closed / Cancelled

Sub-issue lifecycle

Open → Under Repair → For Inspection → For Confirmation → Done (or Deferred)

The five phases

- 1 **Create & assign (Admin).** Admin opens a ticket for a vehicle and assigns a Custodian to inspect it.
- 2 **Inspect (Custodian).** The Custodian physically inspects and records the sub-issues found. Confirming real problems pulls the vehicle Under Maintenance immediately.
- 3 **Dispatch & repair (Admin → Maintenance).** Admin assigns a mechanic to each sub-issue; the mechanic performs the work and logs it — parts used, cost, an optional photo/document, and schedule notes.
- 4 **Verify & confirm (Custodian → Admin).** The Custodian runs a functional test (§7); on pass, the Admin gives the final confirmation and the sub-issue becomes Done.
- 5 **Close (Admin).** Once every sub-issue is resolved, the Admin closes the ticket. Closing is the single action that can return the vehicle to Available.

Single choke point. Only closing a ticket (or cancel / uncancel / delete / decommission) may change a vehicle's availability. Confirming one sub-issue never frees the vehicle on its own — the vehicle only returns to service once *none* of its tickets are still open, so a second unrelated ticket can never be silently ignored.

7. Verification — the functional test

Verification is not a rubber stamp. When a repair is submitted, the **Custodian** — deliberately a different role from the mechanic who did the work ("don't grade your own homework") — must physically operate the vehicle against a **type-specific checklist** and attest to it.

Vehicle type	Example checklist items
Any	Engine/power starts normally · no warning indicators or abnormal noise · the reported problem no longer occurs
Land	Brakes respond properly · completed a short test drive
Fire truck	Water pump reaches full pressure

Water (rescue boat)	Engine runs under load on the water · bilge pump operates · no hull leaks
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- **Any failed check** → **Rejected**, and the sub-issue bounces straight back to the mechanic to redo — exactly like a failed UAT returning to development.
- **All checks pass** → the tester must tick an *attestation* ("I personally operated and tested this vehicle") before it can be Approved.
- The completed checklist is recorded and shown to the Admin before final confirmation, so a repair reaches "Done" because it was *proven to work*, not because a button was clicked.

8. Deferred sub-issues & decision-close

Real life: one sub-issue sometimes cannot be finished (no budget, part on back-order). A ticket must not be trapped forever by a single unfixable line item.

- **Deferred** is a second, terminal way for a sub-issue to end — a *recorded decision not to fix it now*, with a mandatory reason and who decided.
- A ticket can close once every sub-issue is **resolved** — Done or Deferred.
- **Decision-close**: an Admin can close a ticket with unfinished work; the leftovers are auto-marked Deferred under one reason.
- **Fit-for-service gate**: when closing with unfinished work, the Admin must answer "*Is the vehicle safe to return to service?*" — **No** keeps the vehicle out of service even though the ticket is closed, so a safety call is never laundered into a checkbox.
- **Breadcrumb**: every deferred defect opens (or re-surfaces) a Pending issue report, so a known-but-unfixed problem is never forgotten.
- **Aging**: each open ticket shows how many days it has been open and is flagged once it passes 30 days.

9. Ticket archive & recovery

- **Closed** tickets are permanently locked and archived — never reopened.
- A ticket **deleted** with no finished work leaves no trace. A ticket deleted after real progress (≥ 1 sub-issue Done) is archived as *Deleted* and can be **reopened**, reconstructing the ticket and all its sub-issues from a saved snapshot — the safety net for an accidental delete.
- **Cancel / Uncancel** lets an Admin pause and restore a ticket that was never completed.

10. Maintenance records, schedules & condition monitoring

- **Maintenance Records** — the unified repair ledger. Every confirmed sub-issue automatically writes a completed record (cost, parts, personnel, verification), and standalone records can be created and run through their own verify/confirm flow.

- **Maintenance Schedule** — preventive maintenance planning per vehicle and date.
- **Condition Monitoring** — recorded condition checks that update a vehicle's physical state.

11. Fleet intelligence

Four capabilities added to make the system reason about the fleet, not just track it.

11.1 Emergency Readiness (coverage)

The dashboard answers the question that actually matters for emergencies: *for each vehicle type, do we have a ready unit right now?* Availability is grouped by type and shown as "**Ambulance: 1 of 1 ready**". The moment the last ready unit of any type goes down, a hard **NO COVERAGE** alarm is raised.

11.2 Preventive maintenance with teeth

Scheduled maintenance is no longer a silent calendar. Any schedule that is **overdue** or **due within 7 days** surfaces on a dashboard **Preventive Watch** list — colour-coded OVERDUE / DUE SOON — so prevention is visible *before* a breakdown, not after.

11.3 Reliability & repeat-failure detection

- **Recurring flag:** when a new ticket repeats a Main Issue that was already fixed-and-closed on the same vehicle within 90 days, the ticket is badged "*Recurring — Nth time*", so a chronic unit is not hidden as unrelated tickets.
- **Reliability lens (per vehicle):** the vehicle profile shows failures in the last 6 and 12 months, lifetime repair spend, average days out of service, and a **chronic** warning (3+ failures in 6 months, or any recurrence).

11.4 Decommission (end-of-life)

A distinct, documented retirement — separate from a temporary archive. An Admin decommissions a vehicle with a **mandatory reason** (beyond economical repair, too old, totalled). A decommissioned vehicle:

- Is removed from readiness/coverage counts and from every "pick a vehicle" list — no new tickets or issues.
- Keeps its full history for the record.
- Requires open tickets to be settled first (use decision-close for an unfinishable one).
- Can be **recommissioned** by an Admin if it was retired by mistake.

12. Dashboard, notifications, reports & audit trail

- **Dashboard** — role-aware metrics, fleet-status donut, Emergency Readiness board, Availability Forecast (who is out and when they return), Preventive Watch, activity chart, and recent updates.
- **Notifications** — every workflow hand-off notifies the right person (inspection assigned, work order dispatched, verification required, repairs approved, ticket ready to close, deferred, closed, reopened...).
- **Reports** — vehicle, issue, maintenance, schedule, location and history reports with date and attribute filters.
- **Activity Logs & Vehicle History** — a full, tamper-evident audit trail of who did what, when, and (with multi-role) under which role.

13. End-to-end example workflows

Workflow A — A fire truck that overheats, fixed and returned to service

- 1 Custodian reports "Engine overheating" on Fire Truck #2 (severity High).
- 2 Admin creates a ticket "Overheating" from the report and assigns the Custodian to inspect.
- 3 Custodian inspects and logs two sub-issues: "Low coolant level" and "Faulty radiator". The truck goes Under Maintenance automatically.
- 4 Admin assigns a mechanic to each; the mechanic refills coolant / replaces the radiator and logs the parts, cost, and a photo.
- 5 Custodian runs the functional test — engine starts, no warnings, *pump reaches full pressure*, problem gone — ticks all Pass, attests, and Approves.
- 6 Admin confirms each sub-issue (2 of 2 Done) and closes the ticket. The truck returns to Available, and Emergency Readiness shows Fire Truck coverage restored.

Workflow B — A repair that can't be finished this quarter (decision-close)

- 1 An ambulance ticket has two sub-issues; the brakes are fixed, but the water-pump seal is on back-order with no budget until Q4.
- 2 Admin defers the pump sub-issue with the reason "seal on back-order, no budget until Q4". A follow-up Pending issue report is opened automatically so it isn't forgotten.
- 3 With every sub-issue now resolved (1 Done, 1 Deferred), the Admin closes the ticket — answering the fit-for-service question. Because the deferred item is not safety-critical, they answer Yes, and the ambulance returns to service; the outstanding work stays visible as its breadcrumb report.

Workflow C — One person, two hats (small barangay)

- 1 Admin creates an account for Juan and ticks both **Custodian** and **Maintenance Personnel**. Juan's sidebar now shows both portals.
- 2 Juan (as Custodian) reports and inspects an issue; the Admin assigns the repair to Juan (as Maintenance); Juan logs the repair; Juan (back as Custodian) runs the functional test and accepts it.
- 3 The audit trail records each step under the exact role Juan wore — reporter, repairer, tester — so accountability is intact even with one person.

Workflow D — Retiring a chronic rescue boat

- 1 The rescue boat's profile shows a **chronic** reliability warning — 4 failures in 6 months and rising lifetime spend.
- 2 Its last unfinishable ticket is decision-closed (fit-for-service: No), so it has no open tickets.
- 3 Admin decommissions it with the reason "beyond economical repair". It leaves the readiness math for good, its history is preserved, and Water-rescue coverage now honestly reads what remains.

14. Reference tables

Vehicle status × condition — is it usable?

Status	Meaning	Dispatchable?
Available	Ready for service (condition may still note a non-blocking flag)	Yes
Under Maintenance	Actively being repaired / pulled from service	No
Inactive	Temporarily archived (reversible)	No
Decommissioned	Permanently retired at end-of-life	No (never)

Sub-issue statuses

Status	Meaning
Open	Logged, awaiting a mechanic
Under Repair	Mechanic is working on it
For Inspection	Repair submitted, awaiting the Custodian's functional test
For Confirmation	Passed the test, awaiting the Admin's final verdict
Done	Fixed, tested, and confirmed

Deferred	A recorded decision not to fix it now (with reason + breadcrumb)
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At a glance — everything the system can do

- Vehicle registry (Land & Water)
- Archive / restore
- Decommission / recommission
- Vehicle types & domains
- Location hubs & map
- Issue reporting
- Main-issue / sub-issue tickets
- Per-sub-issue mechanic dispatch
- Repair logging (parts, cost, photo)
- Type-specific functional test
- Two-tier verify & confirm
- Deferred + decision-close
- Fit-for-service gate
- Ticket aging
- Recurring-failure flag
- Ticket archive & reopen
- Maintenance ledger
- Preventive schedules + watch
- Condition monitoring
- Emergency readiness board
- Per-vehicle reliability lens
- Multi-role accounts
- Role-aware dashboard
- Notifications
- Reports
- Activity & vehicle history audit

Barangay Vehicle Management System · Version 2.0.3 · This document reflects the system as built and tested (backend feature suite: 67 automated tests passing).